

Platelets in atherosclerosis.

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Beyond obvious functions in haemostasis and thrombosis, platelets are considered to be essential in proinflammatory surroundings such as atherosclerosis, allergy, rheumatoid arthritis and even cancer. In atherosclerosis, platelets facilitate the recruitment of inflammatory cells towards the lesion sites and release a plethora of inflammatory mediators, thereby enriching and boosting the inflammatory milieu. Platelets do so by interacting with endothelial cells, circulating leukocytes (monocytes, neutrophils, dendritic cells, T-cells) and progenitor cells. This cross-talk enforces leukocyte activation, adhesion and transmigration. Furthermore, platelets are known to function in innate host defense through the release of antimicrobial peptides and the expression of pattern recognition receptors. In severe sepsis, platelets are able to trigger the formation of neutrophil extracellular traps (NETs), which bind and clear pathogens. The present antiplatelet therapies that target key pathways of platelet activation and aggregation therefore hold the potential to modulate platelet-derived immune functions by reducing cellular interactions of platelets with other immune components and by reducing the secretion of inflammatory proteins into the milieu. The objective of this review is to update and discuss the current perceptions of the platelet immune constituents and their prospect as therapeutic targets in an atherosclerotic setting.